

Designing a Competitive Digital Platform for Verified Long - Term Environmental Impact Allocation



Aeterna

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Introduction

Over the last decade, environmental impact markets have expanded significantly as corporations attempt to meet increasingly ambitious sustainability commitments. Firms are under growing pressure from regulators, investors, and consumers to demonstrate credible environmental performance through Environmental, Social and Governance (ESG) strategies and net-zero targets.

Despite this growth, environmental credit markets remain deeply controversial. Several recent investigations have revealed that many carbon offset projects deliver limited or temporary ecological benefits. In numerous cases, projects fail to guarantee long-term carbon sequestration or biodiversity restoration, leading to widespread accusations of greenwashing.

This situation reflects a deeper economic problem. Environmental markets suffer from several structural distortions that prevent efficient allocation of capital toward projects generating durable ecological value.

Three distortions are particularly important.

First, information asymmetry prevents buyers from observing the true ecological quality of environmental projects.

Second, moral hazard emerges after contracts are signed, as project providers may reduce long-term maintenance effort.

Third, behavioral biases, particularly hyperbolic discounting, cause firms to systematically undervalue projects that generate benefits over long time horizons.

Together, these distortions create a persistent underinvestment in long-term ecological assets.

This report proposes the design of Aeterna, a digital multi-sided platform dedicated to allocating high-integrity long-term environmental impact projects.

Unlike conventional carbon marketplaces, Aeterna focuses exclusively on projects that generate verified environmental benefits over a minimum horizon of twenty years.

The central research question guiding this project is therefore:

How can a digital multi-sided platform correct information asymmetries and behavioral biases in environmental markets while using incentive-compatible mechanisms to allocate long-term verified environmental impact efficiently?

Market Structure and Stakeholders

Environmental impact markets involve several actors with distinct incentives and informational constraints.

Project Providers (Supply Side)

The supply side of the platform consists of organizations capable of implementing environmental restoration projects.

These include:

- regenerative farmers implementing soil carbon sequestration practices
- environmental NGOs conducting forest restoration programs
- local governments restoring wetlands or biodiversity corridors
- landowners managing conservation areas

These actors generate environmental benefits such as carbon sequestration, biodiversity recovery, and ecosystem resilience.

However, these projects typically require significant upfront investment and long-term ecological monitoring, which creates financing constraints.

Corporate Buyers (Demand Side)

The demand side consists of companies seeking credible environmental impact to meet sustainability commitments.

Typical buyers include:

- airlines
- technology firms
- manufacturing companies
- energy corporations

These firms purchase environmental credits for several purposes:

- achieving net-zero commitments
- improving ESG ratings
- reducing reputational risks
- signaling environmental responsibility to stakeholders

However, buyers face strong uncertainty regarding the true ecological quality and durability of projects.

The platform (Aeterna)

Aeterna functions as a digital intermediary connecting project providers and corporate buyers.

The platform performs several essential functions:

- environmental project verification
- project scoring and ranking
- Allocation through auction mechanisms
- secure payment escrow systems
- long-term monitoring of ecological performance

In economic terms, Aeterna acts as a trust infrastructure designed to reduce transaction costs and improve information transparency in environmental markets.

Information Asymmetry and Market Failure

Environmental impact markets are strongly affected by information asymmetry, a situation in which one party in a transaction possesses more information than the other.

In this market:

- project providers know the true ecological quality of their projects
- corporate buyers cannot easily verify whether the promised environmental impact will occur

This asymmetry generates two classical economic distortions.

Adverse Selection

Adverse selection occurs when buyers cannot distinguish between high-quality and low-quality projects.

This dynamic corresponds to the “Market for Lemons” described by George Akerlof.

When project quality is unobservable, buyers rationally offer a price reflecting average expected quality. High-quality providers cannot recover their higher costs and may exit the market.

As a result, low-quality projects dominate the market and overall credibility collapses.

Moral Hazard

Moral hazard arises after a contract has been signed.

Once project providers receive funding, their long-term actions become difficult to monitor.

Without continuous verification, providers may reduce ecological maintenance effort.

This risk is particularly severe in projects with long time horizons, such as forest restoration programs requiring decades of monitoring.

Behavioral Biases in Environmental Investments

In addition to informational distortions, environmental investment decisions are influenced by behavioral biases.

The most significant bias affecting corporate buyers is hyperbolic discounting.

Hyperbolic discounting describes the tendency for decision-makers to heavily discount future benefits relative to immediate outcomes.

In environmental markets, this bias leads companies to prefer projects that generate immediate reputational visibility rather than projects delivering stronger long-term ecological benefits.

This dynamic can be formalized through the following discounting function:

$$U_t = \frac{V}{1 + kt}$$

Where:

U_t represents the perceived value of a future benefit

V represents the true long-term value

t represents time

k represents the present-bias parameter

Because firms discount future benefits excessively, projects generating durable environmental impact are systematically undervalued.

Data Mobilization and Environmental Verification

To correct information asymmetry, Aeterna integrates environmental monitoring technologies that transform hidden ecological quality into observable signals.

These technologies include:

- satellite monitoring of forest biomass
- soil carbon measurement technologies
- biodiversity monitoring indicators

Satellite systems such as ESA Sentinel-2 and NASA GEDI provide continuous observation of vegetation growth and forest cover.

The integration of environmental data serves three economic functions:

1. reducing hidden information
2. improving credibility of ecological signals
3. enabling performance-based environmental contracts

In this sense, data functions as an institutional mechanism that restores trust in environmental markets.

Mechanism Design: The Aeterna Hybrid Allocation System

The core innovation of Aeterna lies in its hybrid allocation mechanism combining project screening and auction allocation.

Stage 1 – Permanence-Weighted Scoring

Each project receives a score calculated as follows:

$$S = \frac{V \times B \times P}{C}$$

Where:

V = verified environmental impact volume

B = biodiversity and ecosystem co-benefits

P = permanence probability (20+ year survival probability)

C = cost per unit of environmental impact

This scoring system converts hidden ecological quality into a transparent signal visible to buyers.

Only projects exceeding a minimum score threshold enter the allocation phase.

Stage 2 – Vickrey Auction

Projects that pass the screening phase are allocated through a second-price sealed-bid auction.

Auction rule :

- buyers submit sealed bids
- the highest bidder wins
- the winner pays the second-highest bid

This mechanism is strategy-proof, meaning bidders maximize payoff by revealing their true valuation.

This ensures efficient allocation of projects to buyers with the highest willingness to pay.

Incentive Alignment: Staged Escrow Payments

To prevent moral hazard, Aeterna introduces a staged payment structure.

Instead of paying the full project value upfront:

- 30% of funds are released at project initiation
- 70% are stored in escrow

Payments are released gradually depending on ecological performance verification.

This aligns project providers' incentives with long-term ecological success.

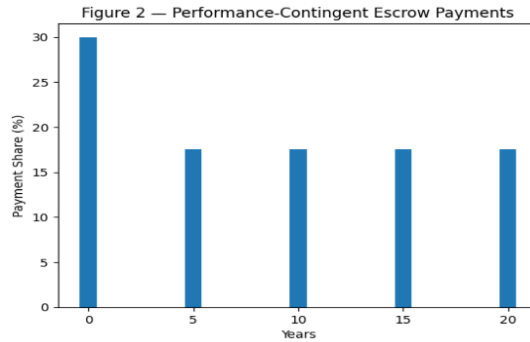


Figure 1 – Performance-Contingent Escrow Payments

Figure 1 compares the relative credibility of several environmental credit platforms. While existing platforms mainly focus on certification or trading infrastructure, Aeterna integrates environmental data verification, incentive-compatible allocation mechanisms, and long-term ecological permanence. This positioning aims to address credibility challenges in current environmental markets.

Behavioral Mechanism Design

Aeterna integrates behavioral insights into platform architecture.

Permanence Badge:

Companies funding long-term projects receive an immediate reputational signal called the Aeterna Platinum Guardian Badge.

This transforms future ecological benefits into immediate reputational capital.

Default Longevity Filter:

the platform interface defaults to projects with minimum 20-year permanence.

Because users tend to follow default options, this design leverages status quo bias to encourage long-term investment.

Platform Strategy and Network Effects

Aeterna operates as a multi-sided digital platform connecting project providers and corporate buyers.

Such platforms benefit from indirect network effects.

As the number of projects increases, the platform becomes more attractive to buyers.

As the number of buyers increases, competition in auctions increases, improving price discovery and revenues for project providers.

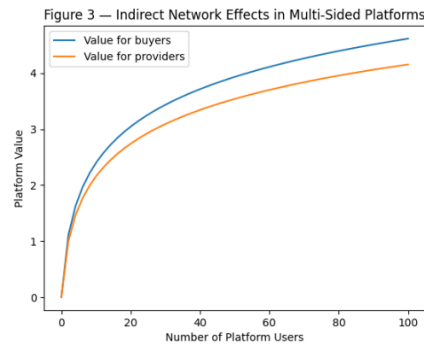


Figure 2 – Indirect Network Effects in a Multi-Sided Platform

Figure 3 illustrates indirect network effects: as the number of users increases on one side of the platform, the value of the platform increases for users on the other side.

Business Model

Aeterna adopts an asymmetric pricing strategy typical of digital platforms.

Supply side participation is free to attract high-quality ecological projects.

Corporate buyers generate revenue through:

- **10% transaction commission**
- premium subscriptions providing real-time environmental monitoring dashboards

In this model, buyers effectively pay for:

- credible environmental verification
- reduced information asymmetry
- transparent monitoring infrastructure

The Aeterna

Business Model Canvas



Competition and Market Positioning

Aeterna competes with existing environmental credit platforms including:

- Gold Standard
- Pachama
- Carbon Trade Exchange

However, most of these platforms focus primarily on carbon credit certification or trading liquidity.

Aeterna differentiates itself by specializing in:

- long-term ecological permanence
- incentive-compatible allocation mechanisms
- integrated environmental data verification

Comparative Analysis: Aeterna vs Existing Environmental Impact Platforms

Criteria	Gold Standard	Pachama	Carbon Trade Exchange		Aeterna (Proposed)
Primary Focus	Certified carbon offset projects	Nature-based carbon projects (forestry)	Carbon credit trading marketplace		20+ Year Verified Biodiversity & Carbon Permanence
Market Type	Certification + marketplace	Curated marketplace	Exchange-style trading platform		Hybrid auction + scoring platform
Time Horizon Emphasis	Mixed (short & medium-term credits)	Mainly medium-term forestry	Mostly short/medium-term tradable credits		Strict long-term permanence (20+ years)
Allocation Mechanism	Fixed-price listing	Fixed-price curated offers	Market trading (spot exchange)		Screening + Vickrey Auction
Incentive Compatibility	Limited	Limited	Market pricing		Strategy-proof auction + staged escrow
Moral Hazard Solution	Third-party verification	Monitoring via remote sensing	Registry-based transfer		Performance-contingent escrow payments
Behavioral Design	Minimal	Minimal	None		Permanence badge + longevity default
Data Integration	Certification-based	AI + satellite monitoring	Registry tracking		Satellite + IoT + dynamic permanence scoring
Pricing Model	Project listing + credit sale	Project brokerage	Transaction-based trading		Asymmetric: subsidized supply + commission on buyers
Competitive Edge	Credibility standard	Tech-enabled forest monitoring	Liquidity & trading		Long-term trust infrastructure + permanence specialization

Conclusion

Environmental impact markets currently suffer from structural inefficiencies caused by information asymmetry, behavioral biases, and weak verification mechanisms.

The Aeterna platform proposes a new institutional architecture combining environmental data verification, mechanism design, and behavioral nudges to allocate environmental capital more efficiently.

By transforming hidden ecological quality into observable signals and aligning incentives over long time horizons, Aeterna strengthens trust and improves capital allocation in environmental markets.

Ultimately, the platform demonstrates how digital technologies and behavioral economics can jointly address environmental market failures while directing investment toward projects that generate durable ecological value.